Python Beam Extraction and Monte Carlo Simulator (PY-BEMCS)

Bharat Singh Rawat

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Abstract

The primary inspiration behind PY-BEMCS originated during my doctoral research, where I worked extensively with ion sources and beam extraction systems. Throughout that time, I evaluated numerous simulation tools that utilized Finite Element Method (FEM) techniques to model beam extraction processes. However, the vast majority of these were proprietary commercial codes locked behind prohibitive license fees and recurring service charges.

My vision for PY-BEMCS is to break down these barriers by providing a powerful, fully open-source beam extraction simulator dedicated to non-commercial research by academics and students. A core design philosophy of this project is accessibility: it is engineered to be as user-friendly as possible, removing the steep learning curve of complex programming so that researchers can focus directly on the physics. As detailed in this document, the resulting engine utilizes a Hybrid Particle-in-Cell (PIC) / fluid approach to model plasma dynamics, coupled with Monte Carlo Collisions (MCC) for neutral interactions, and a finite-difference thermal model for structural degradation.

I officially initiated this project today (26/03/2026), on my 34th birthday, as a personal commitment to fostering accessible, open science for the next generation of researchers. This document details the physical models, numerical methods, and multiphysics couplings implemented in PY-BEMCS. The engine utilizes a Hybrid Particle-in-Cell (PIC) / fluid approach to model plasma dynamics, coupled with Monte Carlo Collisions (MCC) for neutral interactions, and a finite-difference thermal model for structural degradation.

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1 Fundamental Domain & Scaling

To simulate nanosecond-scale plasma physics alongside minute-scale thermal and erosion dynamics, the engine employs several numerical scaling techniques:

- **Macro-Particles:** Instead of tracking billions of individual ions, the engine tracks "macro-particles." Each macro-particle represents a specific number of real ions, defined by the weight W. This weight is dynamically auto-scaled based on the plasma density (n_0) to maintain an optimal particles-per-cell ratio.

- **Artificial Electron Mass:** Tracking true electrons requires femtosecond timesteps due to their high velocity. To allow for a larger standard timestep ($\Delta t = 10^{-9}$ s) while still resolving electron backstreaming trajectories, the explicitly tracked secondary electrons are assigned an artificial mass ($m_e = m_{Xe}/100$).

- **Thermal Acceleration Factor:** Thermal dynamics happen millions of times slower than ion transits. A multiplier (a_{th}) scales the specific heat and radiative cooling equations so that grid temperatures and deformations can be observed in a reasonable number of simulation iterations.

2 Electrostatics & Space Charge (Hybrid PIC)

2.1 The Non-Linear Poisson Equation

The electric potential $V(x, y)$ is solved on a 2D Cartesian grid. Instead of tracking bulk plasma electrons as particles, the engine treats them as a massless fluid following the Boltzmann relation. The governing Poisson equation is:

$$\nabla^2 V = -\frac{\rho_{total}}{\epsilon_0} = -\frac{\rho_{ion} + \rho_e}{\epsilon_0} \tag{1}$$

Where ρ_{ion} is gathered directly from the positions of the explicitly simulated macro-ions using nearest-grid-point weighting. The electron charge density ρ_e assumes a Maxwell-Boltzmann distribution referencing the upstream plasma potential (V_{plasma}) and electron temperature ($T_{e,up}$):

$$\rho_e(V) = -qn_0 \exp\left(\frac{\min(V, V_{plasma}) - V_{plasma}}{T_{e,up}}\right) \tag{2}$$

2.2 Numerical Solver

Because the boundaries change dynamically during erosion but remain static for hundreds of steps at a time, the engine pre-computes the LU Factorization of the Laplacian sparse matrix. To handle the non-linearity of the Boltzmann electrons, the solver uses Picard Iterations with successive under-relaxation ($\omega = 0.2$) to ensure the potential field converges smoothly without numerical oscillation.

3 Ion Kinematics & Injection

3.1 Bohm Criterion Injection

Xenon ions (Xe^+) are injected at the upstream boundary. To satisfy the physical requirements of sheath formation, ions must enter the simulation domain at or above the Bohm velocity:

$$v_{Bohm} = \sqrt{\frac{qT_{e,up}}{m_{Xe}}} \tag{3}$$

The total ion current I_{ion} entering the domain is scaled by the effective injection area and the pre-sheath density drop coefficient (0.61):

$$I_{ion} = 0.61 \cdot q \cdot n_0 \cdot v_{Bohm} \cdot A_{inject} \tag{4}$$

3.2 Particle Push

Particles are accelerated by the local electric field $\vec{E} = -\nabla V$. The field is interpolated continuously from the discrete grid to the exact particle coordinates.

The kinematic update uses an explicit integration scheme:

$$\vec{v}(t+\Delta t) = \vec{v}(t) + \frac{q}{m}\vec{E}\Delta t \tag{5}$$

$$\vec{r}(t+\Delta t) = \vec{r}(t) + \vec{v}(t+\Delta t)\Delta t \tag{6}$$

4 Charge-Exchange (CEX) Collisions

Fast beam ions can interact with slow background neutral Xenon gas, exchanging an electron. The relative velocity g between the ion and neutral is approximated as the ion's magnitude of velocity. The empirical cross-section σ_{CEX} for Xe^+ on Xe is calculated as:

$$\sigma_{CEX}(g) = \left(-0.8821\ln(g) + 15.1262\right)^2 \times 10^{-20}\ [\mathrm{m}^2] \tag{7}$$

The probability P_{CEX} of a collision occurring for any given macro-ion in a single timestep Δt is evaluated using the Monte Carlo method:

$$P_{CEX} = 1 - \exp\left(-n_n \sigma_{CEX}(g) g \Delta t\right) \tag{8}$$

If a collision occurs probabilistically, the ion's velocity is instantly reassigned to match the thermal velocity of the background neutral gas.

5 Grid Erosion & Structural Failure

5.1 Sputtering Yield

Physical damage is quantified by an empirical sputter yield $Y(E)$, representing Molybdenum atoms removed per incident Xenon ion. The threshold energy for sputtering is roughly 30 eV. The yield curve is modeled as:

$$Y(E) = \begin{cases} 1.05 \times 10^{-4}\,(E-30)^{1.5} & E > 30 \text{ eV} \\ 0 & E \leq 30 \text{ eV} \end{cases} \tag{9}$$

5.2 In-Situ Remeshing

Damage is accumulated in a 2D scalar field. When a grid cell's accumulated damage exceeds the defined structural threshold, the cell is flagged as broken. The engine immediately removes this cell from the solid boundary mask, rebuilding and re-factorizing the Sparse Laplacian matrix to allow plasma to flow through the newly eroded hole.

6 Secondary Electron Emission (SEE)

High-energy ion impacts on the grids can eject electrons from the metal lattice. The empirical yield γ_{SEE} (electrons emitted per incident ion) is modeled linearly

above a baseline:

$$\gamma_{SEE}(E) = \max\left(0, \min\left(1, 0.05 + 10^{-4}E\right)\right) \tag{10}$$

If an emission event occurs, macro-electrons are spawned slightly offset from the grid wall with a random thermal energy of roughly 2.0 eV. They are then explicitly tracked by the PIC engine.

7 Thermal Multiphysics

7.1 Kinetic Heating

Kinetic energy from impacting macro-particles is deposited into the grid geometry as heat. The thermal energy ΔQ deposited per timestep is calculated from the Xenon mass and the macro-weight W:

$$\Delta Q = \frac{1}{2} m_{Xe} |\vec{v}|^2 W \tag{11}$$

This energy increases the local cell temperature based on the volumetric heat capacity C_{cell}:

$$\Delta T_{heat} = \left(\frac{\Delta Q}{C_{cell}}\right) a_{th} \tag{12}$$

7.2 Radiative Cooling

Grids dissipate heat into the vacuum via Stefan-Boltzmann radiation. The temperature reduction per timestep for boundary cells is:

$$\Delta T_{cool} = \frac{\epsilon \sigma_{SB} A_{cell}}{C_{cell}} \left(T^4 - 300^4\right) \Delta t \cdot a_{th} \tag{13}$$

7.3 Thermal Expansion

As the accelerator grid temperature (T_a) rises above the 300K baseline, linear thermal expansion alters the physical gap distance d_{gap} between the screen and accelerator grids.

$$\Delta d_{gap} = \alpha_{moly} \cdot d_{gap} \cdot (T_a - 300) \tag{14}$$

If the gap distance changes by more than 0.05 mm, the engine recalculates the mesh geometry to capture the effects of thermal grid warping.

8 Physics Validation and Benchmarks

To ensure the physical accuracy of the simulated ion optics and multiphysics couplings, PY-BEMCS includes a suite of automated, multi-core benchmarking scripts. These tests validate the core solvers against established theoretical models, empirical beam extraction behaviors, and known facility effects.

8.1 Beam Divergence vs. Perveance

Objective: To validate the core ion kinematics, space-charge repulsion, and beam divergence of the grid apertures.

Physical Principle: The focal length and trajectory of the extracted beamlets are governed by the perveance (P), defined as:

$$P = \frac{I_b}{V_T^{3/2}} \tag{15}$$

where I_b is the beamlet current and V_T is the total extraction voltage. According to Child-Langmuir and thin-lens theory, there is an optimum perveance where the upstream plasma sheath is perfectly flat, resulting in minimum beam divergence. If P is too high, the sheath bulges outward and space-charge forces cause the beam to diverge. If P is too low, the sheath becomes deeply concave, causing the beam to over-focus and cross over itself.

Simulation Setup: The benchmark isolates the pure optics by disabling the downstream neutralizer model. It sweeps the upstream plasma density (n_0) to systematically vary the perveance across different grid gap lengths. The engine tracks the 95th percentile divergence angle of ions exiting the domain. The output successfully reproduces the classic "U-Curve" characteristic of dual-grid ion optics.

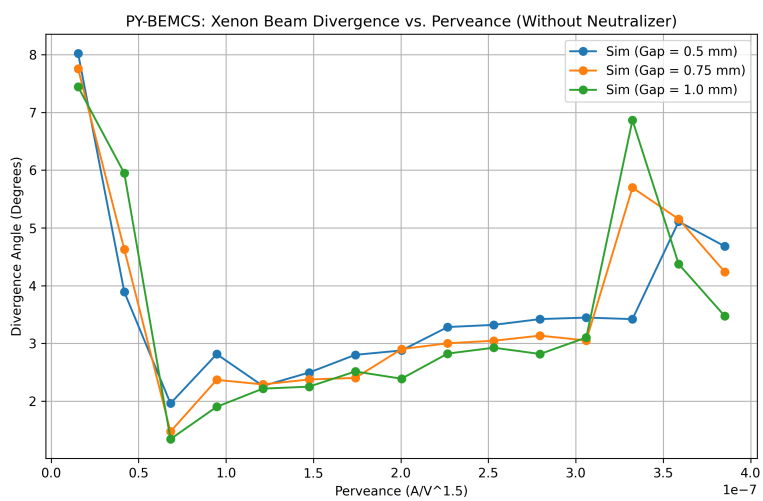

Figure 1: Simulated beam divergence for three grid gaps.

The Fig. 1 shows the perveance benchmark results for three different grid gaps d=0.5mm,0.75mm and 1mm. The other parameters are V_a=1650V, V_s=-350V, t_s,t_a=1mm,n_0=0,$n_{\text{plasma}} = 1 \times 10^{16}$ to $2.5 \times 10^{17}\,\text{m}^{-3}$. As can be seen

from the curve the Divergence Vs Perveance characterstics follow the typical U shaped curve. However, at higher perveances than 3×10^{-7}, the divergence drops again; this can be due to the ions crashing into the acceleration grid.

8.2 Electron Backstreaming (EBS) Limit Test

Objective: To validate the 2D Poisson solver and its ability to accurately resolve the potential saddle point within the grid apertures.

Physical Principle: In dual-grid systems, the accelerator grid is biased negatively to prevent electrons from the downstream neutralizer plasma from backstreaming into the discharge chamber. The minimum electrostatic potential along the aperture centerline (the saddle point) must remain sufficiently negative (typically ~ -5 V to -10 V) to reflect these electrons. If the accelerator grid voltage (V_a) is too close to zero, this potential barrier collapses.

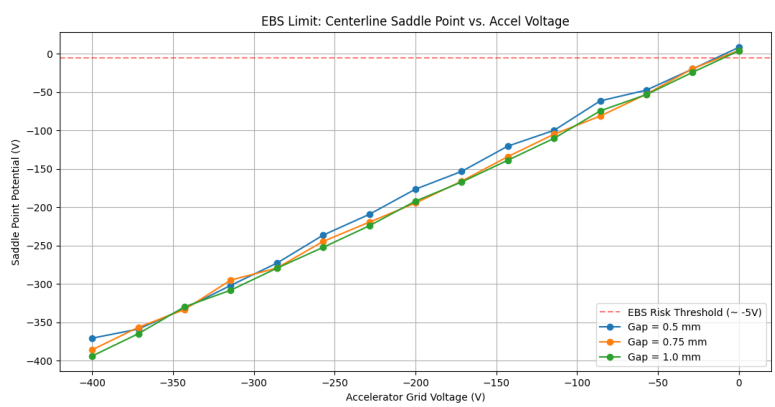

Figure 2: Results for Electron Backstreaming Limit benchmark tests.

Simulation Setup: The benchmark sweeps V_a from a safe operating margin (e.g., -400 V) toward 0 V while keeping the beam current constant. The engine tracks the centerline saddle point potential over time. The resulting curve demonstrates the non-linear collapse of the potential barrier, identifying the exact EBS failure threshold for a specific grid geometry.

8.3 Impingement Current (V-Curve) Benchmark

Objective: To validate particle-to-grid collision mapping, beam containment, and the calculation of grid transparency.

Physical Principle: When the beam operates far from the optimal perveance (either over-focusing or under-focusing), the outer edges of the beamlet physically crash into the accelerator grid geometry. These impacting primary ions are

recorded by the engine as impingement current (I_{imp}). This telemetry allows for the calculation of the grid's effective ion transparency (η), defined as:

$$\eta = 1 - \frac{I_{imp}}{I_{beam}} \tag{16}$$

where I_{beam} is the total extracted beamlet current. High impingement current indicates poor beam optics and leads to rapid structural erosion.

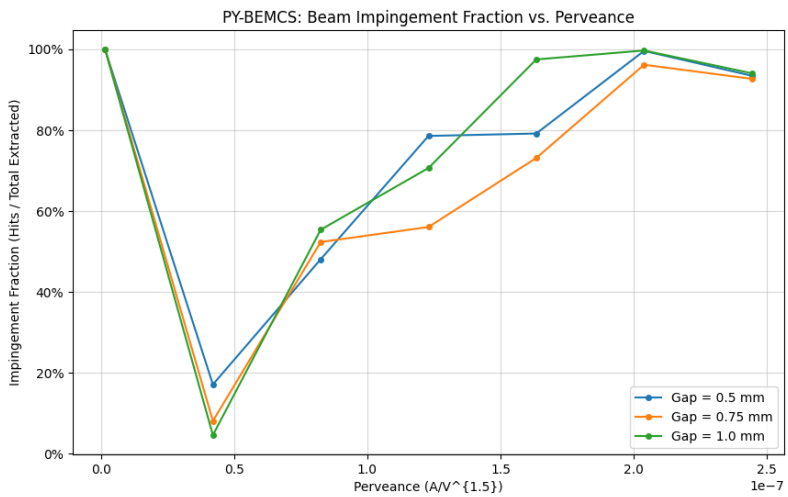

Figure 3: Results for beam impingement benchmark tests.

Simulation Setup: Operating similarly to the divergence sweep, this test varies the perveance but records the steady-state impingement current absorbed by the accelerator grid mask. Because the impingement current spans several orders of magnitude, it is plotted logarithmically. The output generates a sharp V-Curve, identifying the exact operating margins where direct grid interception is minimized and optical transparency is maximized.

8.4 CEX Erosion vs. Background Pressure Benchmark

Objective: To validate the Monte Carlo Collision (MCC) probability algorithms and facility-effect scaling.

Physical Principle: Charge-Exchange (CEX) collision frequency is dependent on the relative velocity of the primary ions, the empirical CEX cross-section, and the density of the background neutral gas (n_n). For a steady-state thruster operating at a fixed voltage and beam current, the production rate of slow CEX ions and consequently the volumetric erosion rate of the accelerator grid should scale linearly with the background chamber pressure. For all the test plots, the

charge exchange ions are assumed to be formed between 1-2mm in between the grids, where the neutral density is the highest; the result may vary if there is a different neutral density distribution . The plasma density is assumed to be around $1 \times 10^{17} \mathrm{m}^{-3}$ for the CEX benchmark.

Simulation Setup: The benchmark locks the optimal perveance and grid voltages while sweeping the background neutral gas density (n_n) to simulate different vacuum facility pressures. The engine integrates the total accumulated damage map on the accelerator grid over the simulation duration. The output validates the MCC engine by demonstrating a perfectly linear relationship between facility background pressure and structural sputter erosion rates.

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9 How to Cite This Model

If you use this PY-BEMCS or its underlying physics engine in your research, please cite the repository as follows:

Plain Text:
Bharat Singh Rawat. (2026). *Python Beam Extraction and Monte Carlo Simulator (PY-BEMCS)* [Source code]. GitHub. https://github.com/Bharat26031992/PY-BEMCS

BibTeX:

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@software{PY-BEMCS,
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  title        = {Python Beam Extraction and Monte Carlo Simulator (PY-BEMCS)},
  year         = {2026},
  publisher    = {GitHub},
  journal      = {GitHub repository},
  howpublished = {\url{https://github.com/Bharat26031992/PY-BEMCS}},
}
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